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Putting cyclists in boxes: An analysis of the bike box

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ABSTRACT

Recent years have seen an uptake in cycling and, with it, an increase in cyclists being involved in crashes. One particularly problematic area is that of urban signalised intersections, which are responsible for a majority of the crashes involving cyclists. One reason is the increased complexity at such locations. The bike box has been hailed as an infrastructure measure to help increase the safety and visibility of cyclists at intersections, most importantly by reducing the risk of blind spot conflicts. Results on the safety benefits are, however, inconclusive. For the bike box to have the desired safety benefits, it is important that cyclists use it as intended. The study presented here uses observations and questionnaire data collected in the Netherlands and Sweden to better understand the use of the bike box. We find that cyclists going straight ahead do not use the bike box as intended, standing to the right-hand side of the cars. Cyclists turning left do use the bike box as intended, but are less common in all the observed locations. This suggests that the bike box can have a positive effect, but the most common users do not position themselves as intended and still risk being involved in blind spot conflicts. Furthermore, this dual use of the bike box can be misinterpreted by drivers, who expect to see cyclists in front, ignoring their presence on the right-hand side. These results can help inform the actual effectiveness of the bike box in increasing the safety of cyclists at signalised intersections.

1. Introduction

Recent years have seen an increased effort to have a modal shift away from private motorised transport to more active modes of travel, such as cycling, since high cycling levels are associated with more sustainable cities (Chou et al., 2023). These sustainability benefits arise since cycling, compared to the use of motorised transport, produces less pollution, relies on fewer non-renewable sources, and is more economical for users (Leister et al., 2018; Meng et al., 2014; Pucher and Buehler, 2008; Rietveld and Daniel, 2004). Despite the benefits of increased cycling levels, the increased exposure also brought an increase in the occurrence of bicycle crashes and fatalities over the last decade in the European Union (European Commission, 2024; Pazdan et al., 2021). The poor safety record of cycling can form a barrier for a further uptake of cycling as a mode of transport (Chen and Shen, 2016; Useche et al., 2019). One of the most common places for crashes involving cyclists to occur is within urban areas and at intersections (Kaplan and Prato, 2013; Werneke and Vollrath, 2014). More than half (57 %) of cyclist fatalities in the EU in 2022 occurred on urban roads, and 32 % of the total number of cyclist crashes resulting in fatalities took place at intersections (European Commission, 2024). Increasing the safety for cyclists at intersections is, therefore, one way to further increase the modal share of cycling and reap more of its benefits.

1.1. Cyclists safety at intersections

Cyclists are at greater risk of being involved in crashes at intersections due to the complex nature of intersections, where a high number of different manoeuvres can take place (Polders et al., 2015; Schröter et al., 2023; Strauss et al., 2013). As intersection size and complexity increase, so does the risk of cyclists being involved in crashes (Saad et al., 2019; Strauss et al., 2013). Schröter et al. (2023) highlight three crash types occurring at intersections to be most relevant for the safety of cyclists: turning crashes, right-of-way crashes, and loss-of-control crashes. Turning crashes between motor vehicles and cyclists at intersections are stated to be the prominent risk to cyclists (Beck et al., 2016; Cantisani et al., 2019; Kaplan and Prato, 2013; Miranda-Moreno et al., 2011; Strauss et al., 2013). These crashes occur when a driver makes a turn and crosses the path of the cyclists continuing straight. One of the reasons for the prevalence of turning crashes are perceptual

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errors, often mentioned as one of the leading factors for crashes at intersections (Rogé et al., 2017; Scarano et al., 2023; Werneke and Vollrath, 2014), due to blind spots conflicts where the driver fails to see the cyclist (Kaplan and Prato, 2013; Thorslund and Lindström, 2020). These types of crashes suggest that treatments aimed at improving the visibility of cyclists at intersections can help to reduce their risk of being involved in a crash. In line with this, DiGioia et al. (2017) argue that for safety treatments to be effective they should address at least one of the following: increase the separation between road users in time and space, increase the visibility of vulnerable road users and the line of sight between road users, reduce the number of interactions between road users, and reduce the speed of motor vehicles.

1.2. The case for the bike box

Typical ways to reduce the risk of cyclists being involved in turning crashes are changes to the geometry of an intersection, changing turning radii, or the introduction of exclusive bicycle signal phases and infrastructure such as bicycle lanes or bike boxes (Strauss et al., 2013). Although inconclusive, Gårder et al. (1994) found that introducing cycle paths on signalised intersections increases the risk of cyclists being involved in a crash. In line with this, Räsänen and Summala (1998) state that although cycle paths increase the safety of cyclists on road sections, they have a negative effect on intersections. Zangenehpour et al. (2016) found that providing cycle tracks reduced safety–critical interactions, but stress that additional intersection treatments would be necessary. Studying five different intersection treatments, Madsen and Lahrmann (2017) find that providing cyclists with their own infrastructure, separated from motorised traffic, provides the greatest safety benefits over on-street bicycle facilities. Separate bicycle infrastructure is, however, not always feasible due to a lack of space. A space-efficient way to increase the visibility of cyclists during the onset of the green phase, provide cyclists with a designated space on the road, reduce their interactions with motorised traffic, and reduce the speed of motor vehicles is the bike box.

The bike box, also referred to as an advanced stop line, consists of an area in front of the stop line for cars where cyclists can position themselves in front of the queue of motor vehicles (de Sande et al., 2022; Weigand, 2008). An example of a bike box is shown in Fig. 1. Having cyclists stand in front of the queue of cars makes them more visible since they are directly in the field of vision of drivers, provides them with a head start during the start of the green phase, and reduces exposure to exhaust fumes (de Lange, 2012; de Sande et al., 2022; Hunter, 2000; Loskorn et al., 2013). The presence of a group of cyclists in front of the car queuing at the traffic light not only increases visibility but can also produce the well-known "safety in numbers" effect (Brüde and Larsson, 1993; Johnsson et al., 2021). In addition to these benefits, having cyclists in front of the queue of cars helps to slow down motorised traffic upon the onset of the green phase as they "block" the path of the cars. Dutch bicycle infrastructure design guidelines state that a bike box is best applied on urban roads under either mixed traffic conditions or with the presence of a bike lane, and in conditions with a relatively high volume of left-turning cyclists or a relatively high volume of cyclists going straight with cars making a right-hand turn (CROW, 2016).



Fig. 1. Example of a bike box (Google, n.d.-b).

1.3. Safety implications of bike boxes

The bike box has been hailed as a measure to reduce the risk of crashes between cyclists continuing straight and right-turning vehicles (Madsen et al., 2022) as well as reducing blind spot incidents for these interactions (de Sande et al., 2022; Fietsersbond, 2022; Schoon et al., 2008), and as a way to reduce stress by providing cyclists with more time during the onset of the green phase (Fietsersbond, 2022). Despite some studies finding a positive effect, the safety effects of the bike box are inconclusive. Whilst Dill et al. (2012) found a reduction in conflicts at intersections where bike boxes were introduced, a study by Madsen et al. (2022), studying conflict rates at intersections in Denmark, did not find conclusive evidence. The authors report a significant decrease in conflict rates at some intersections but a significant increase at other locations, as well as an increase in conflicts between left-turning vehicles and cyclists at one of their study sites. It is worth pointing out that the time between the introduction of the bike box and the study period for the study by Dill et al. (2012) was rather short. This means that it is possible that the positive effect on conflict rates is a result of the new situation rather than the bike box itself, lacking long-term effects and behavioural adaptation. A recent study by Deliali et al. (2023) developed a crash prediction model for several intersection designs and found that bike boxes increased the crash frequency when their effect was isolated by potential confounding factors. The authors do, however, state that their results are not a definitive answer to the safety effect of bike boxes and more research is needed. Studying different intersection treatments, Russo et al. (2023) found that bike boxes have the potential to reduce conflicts between right-turning motor vehicles and cyclists, but only under some conditions. A before-and-after study in Denmark by Buch and Jensen (2017) did not find any statistically significant changes in crash frequencies when bike boxes were in place. These findings indicate that whilst the bike box is used as a treatment to increase the safety of cyclists at signalised intersections, the hypothesised safety benefits are not always realised. It is worth noting that the bike box is designed to increase the visibility of cyclists while waiting at a red light, during the green phase the bike box loses its function and cyclists are mixed with motorised traffic again (de Sande et al., 2022), which can help to explain the mixed safety benefits reported in the literature.

1.4. Use of the bike box

For the bike box to be effective at improving the visibility of cyclists and slowing down cars during the onset of the green phase it is important that cyclists position themselves correctly in the bike box, i.e., in front of the queue of cars, and take up a dominant space on the road. When correctly using the bike box, according to the design philosophy, cyclists who intend to make a left turn move to the left-hand side of the bike box, clearly indicating their intention and taking up space in front of the cars. Cyclists continuing straight take up a position in the middle of the bike box, also positioning themselves in front of the queue of cars, leaving space for cyclists arriving later to enter the bike box as well as for cyclists making a right-hand turn. In doing so, the cyclists waiting at the traffic light form a group in front of the car traffic, avoiding blind spot conflicts and forcing cars to stay behind the cyclists when the signal turns green. This is, however, the envisioned use and relies on cyclists making an effort to adapt their behaviour.

Several studies investigated the use of the bike box based on where cyclists position themselves in the bike box. A questionnaire study in the Netherlands by Schoon et al. (2008) reported that cyclists stated to stand

in front of vehicles in about 60 % of the cases with the second largest group, about 40 %, standing to the right-hand side of a vehicle. Observing three intersections in Germany, Kutusow (2022) found that the positioning of cyclists in a bike box is dependent on their turning manoeuvre. Cyclists going straight were found to spread out across the bike box, with a tendency to position themselves at the right-hand side of the bike box. Left-turning cyclists were also spread out across the bike box with an emphasis on the middle and left-hand side of the bike box. Cyclists turning right position themselves mostly on the right-hand side of the bike box. These findings suggest that the bike box was used as intended, with cyclists taking up a position in front of the queue of cars and filling up the bike box. Observing the behaviour of cyclists in Canada, Casello et al. (2017) found that compliance with the intended use of the infrastructure increased when bike boxes were in place, resulting in fewer illegal crossings by cyclists. This behaviour was found to be independent of traffic volumes.

Such behaviour is, however, not always present. A study on the effect of bike boxes in Oregon found that 84 % of respondents understood the purpose of a bike box, but the observed proportion of cyclists positioning themselves in front of the queue was found to be low (Dill et al., 2012). A similar observation is reported by Madsen et al. (2022), who studied the use of bike boxes in Denmark and by Scott-Deeter et al. (2023), who studied the behaviour of cyclists in a simulated environment. This behaviour is hypothesized to be related to cyclists feeling unsafe being in front of cars (de Lange, 2012), situations where the bike box or approach lane is blocked by cyclists (de Lange, 2012; Madsen et al., 2022; Renčelj et al., 2019), and infrastructure characteristics such as the presence of a continuous bike lane at the opposite side of the intersection (Christofa et al., 2023; Renčelj et al., 2019) or whether cyclist have their own signal phase (Renčelj et al., 2019). Studying where cyclists wait in the bike box in Boston and Somerville (Massachusetts), Christofa et al. (2023) found a sizable portion of cyclists who turn left to wait close to the curb and in the approach lane rather than in the bike box. Such behaviour is clearly not in line with the intention of the bike box, as it still requires cyclists to cross the lane for cars when the signal turns green.

1.5. Aim of this study

The findings related to the safety benefits of the bike box reported across different settings suggest that there are reasons to believe that a bike box does not necessarily lead to a safer situation. This could lead to a false sense of safety when deciding to implement a bike box at an intersection. One reason for this could be that the expected behavioural response to the bike box, i.e., cyclists standing in front of the cars, is not the actual behavioural response by cyclists. Despite their popularity, relatively little work has been done to understand the actual use of the bike box and the reasoning of cyclists to choose a waiting position. Furthermore, the studies that have been conducted come with some important drawbacks. The study by Madsen et al. (2022), for example, took place in situations where cyclists had to make a staggered left turn and the bike box could not be used as intended. The work done by Dill et al. (2012) took place at an intersection where bike boxes were recently introduced, thus potentially capturing a novelty effect. Furthermore, their work took place in the United States, where cycling is less common, which could potentially affect the way cyclists position themselves in the bike box. Whilst the study by Schoon et al. (2008) reports that cyclists stand in front of the queue of cars, their questionnaire design did not consider the presence of other cyclists in the bike box. The aim of the study presented here is, therefore, to better understand the actual positioning of cyclists in the bike box in situations where cyclists are not required to make a staggered left turn, in conditions where bike boxes are common, and where other cyclists are present in the bike box.

Considering them as a safety intervention, having a deeper understanding of the actual behaviour of cyclists shines a light on the use of the bike box and whether the design intentions are actually realised. The latter aspect here is vital since we risk providing planners with a solution that does not work as intended when cyclists do not behave in line with the rationale behind the bike box. This will result in planners unknowingly creating a false sense of safety when committing to the implementation of the bike box at an intersection, rather than providing a real safety benefit for cyclists. Shining light on the use of the bike box will help towards a more evidence-based understanding of the potential safety benefits of the bike box, a necessary condition to make real safety effects, as emphasised by Dupont et al. (2012) and Hauer (2019). It is worth noting that the study presented here focuses on where cyclists position themselves in the bike box with an emphasis on why these choices are made; as such, we do not focus on the direct safety impact of this behaviour here.

2. Data collection and method

For this study, we make use of video observations and a questionnaire study in Sweden and the Netherlands. Two countries with a rich cycling history, and where the bike box is frequently implemented at intersections. While we performed the data collection in two countries, the study presented here is not comparative in nature. The two countries were chosen based on data availability. The video observations are used to gain insights into the actual behaviour of cyclists. The questionnaires, on the other hand, are used to better understand the motivation of cyclists regarding where they stand in the bike box.

2.1. Observations on the use of the bike box by cyclists

To study where cyclists stand in the bike box, we make use of video observations at three intersections in the Netherlands (intersections 1–3) and one in Sweden (intersection 4). The observations took place during either the morning or evening peak hours. The design characteristics of the intersections are shown in Table 1 and an aerial view in Fig. 2. Dutch design guidelines for a bike box state a depth between four and five meters and a minimum width of 2.75 m (CROW Kennisplatform, 2011); Swedish guidelines do not specify the measurements for a bike box (Trafikverket, 2024). All but intersection 2 comply with the Dutch guidelines, but the depth of the bike box is relatively comparable across all locations. The width is larger at intersection 4 as it spans two lanes of traffic instead of one. All four intersections have bicycle lanes leading up to the intersections, and two of the intersections have marked bicycle lanes across the intersection. None of the intersections had a separate signal phase for cyclists.

The total number of green phases and cyclists waiting in the bike box

Table 1
Design characteristics of observed locations.

Location	Intersection width (m)	Intersection across (m)	Bike box width (m)	Bike box depth (m)	Width approach lane (m)	Continuous bike lane?	Separate bicycle signal
Intersection 1	9	25	5	4	2	Yes	No
Intersection 2	32	26	5	3	2	No	No
Intersection 3	8.5	15	4.5	4.5	1.5	Yes	No
Intersection 4	26	38	7.4	4.7	2	No	No

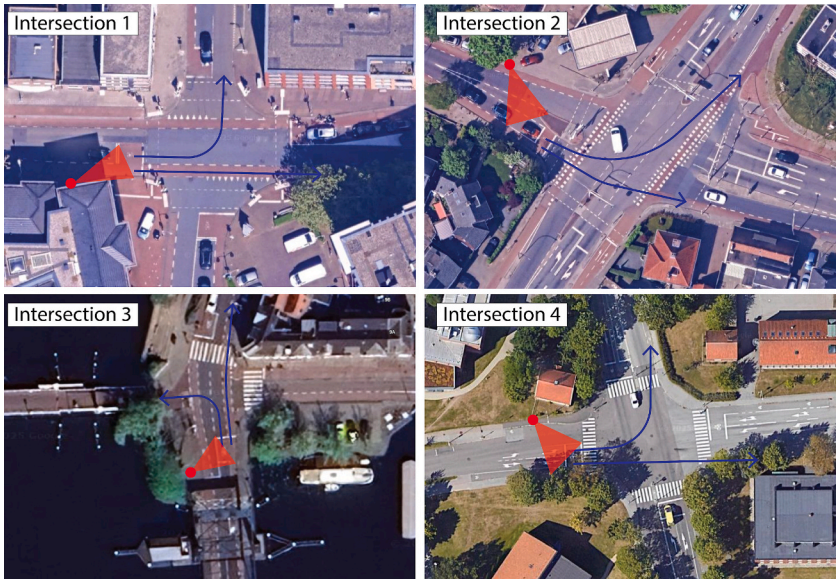


Fig. 2. Overview of the different intersections with the camera angle shown in orange and observed turning directions by cyclists shown by the blue arrows (image source: (Google, n.d.)).

Table 2
Number of green phases observed and total, mean, and maximum number of cyclists waiting in the bike box registered per location.

Location	Number of green phases observed	Number of cyclists registered	Mean number of cyclists in bike box	Max. number of cyclists in bike box
Intersection 1	168	105	2	4
Intersection 2	47	54	2	6
Intersection 3	81	73	2	7
Intersection 4	252	122	1	3
Total	548	232		

registered at each intersection is shown in Table 2. The location where cyclists wait was recorded using the diagram shown in Fig. 3. Cyclists were only registered when stationary during the red phase of the signal. Cyclists who arrived during the green phase and or did not come to a complete stop were excluded. The registration was limited to cyclists continuing straight and those who made a left turn, since it is legal for cyclists to make a right turn on red at intersections in the Netherlands and these cyclists will thus not come to a complete stop in the bike box. The order in which the cyclists arrived was recorded as well, and qualitative assessments of the behaviour shown by cyclists and car drivers were made by the observer.

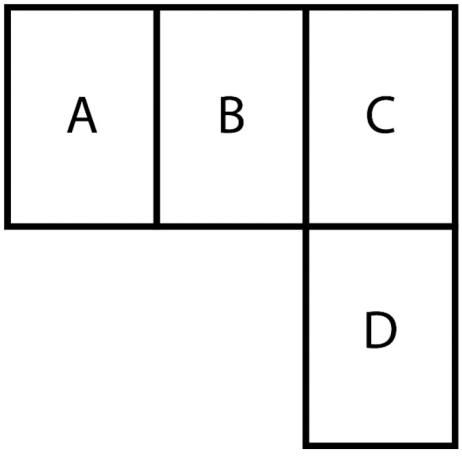


Fig. 3. Registration of the position of cyclists in the bike box.

2.2. Questionnaire study

The aim of the questionnaire study is to understand where cyclists would position themselves based on their intended manoeuvre, going straight or turning left, and the presence of bicycles in the bike box and the motivations for this choice. The questionnaire was distributed in the Netherlands and Sweden, mostly among students and university staff. A total of 154 responses were collected (90 from the Netherlands and 64 from Sweden). Two responses, one from each country, were excluded since the open-ended answers made it clear the respondent assumed the questionnaire was related to where a car driver would wait. This left 152 responses in total, with 89 from the Netherlands and 63 from Sweden.

Participants were shown three different scenarios, shown in Fig. 4, with the difference being the number of bicycles present in the bike box. The images were chosen since they represent common layouts of bike boxes in both countries. It is worth noting that the bike box in the Swedish case spans across two lanes of traffic, given the presence of a dedicated left-turn lane. Such a lane is not present in the Dutch case, where all turning manoeuvres are made from a single lane. In practice, this means that cyclists turning left would not interact with cars continuing straight in the Swedish case. This type of interaction does take place in the design in the Netherlands, without a dedicated lane for traffic turning left. Participants were asked to state in which zone they would wait, A through D, and answer an open-ended question to explain their choice. Participants were also asked about the purpose of the bike box and whether they feel safe using the bike box. When answering no to the latter, they were prompted to explain why. The answers regarding the choice of waiting zones were analysed by comparing the frequency of the chosen zones under different conditions using the Fisher Exact test. The Fisher Exact test is more appropriate than the Chi-Square test here since we deal with a small number of observations in some groups



Fig. 4. Scenes shown in the questionnaire in the Netherlands (top) and Sweden (bottom).

(Bower, 2003; Kim, 2017). The significance levels were corrected using a Bonferroni correction for the pairwise comparisons to control for a potential familywise error rate for multiple comparisons, as suggested by McDonald (2014). Cramer's V effect sizes, the effect size measure associated with the Fisher exact test for larger contingency tables, are reported alongside the statistical testing to provide an insight into the practical relevance of any significant differences (Brydges, 2019). For the interpretation we make use of the benchmarks provided by Cohen (1988). Zero-sum columns (i.e., a zone that was never used) were excluded when calculating the effect size as they cause the calculation to fail. Dropping these columns does not affect the inference of the p-values from the statistical tests, as they add no value to the observed differences. The open-ended questions were subjected to a thematic analysis following the guidelines provided by Braun and Clarke (2006), focusing on keywords and key themes related to feelings of safety and behaviour in mixed traffic conditions.

3. Results

3.1. Video observations

3.1.1. Observed waiting positions

Heatmaps showing the waiting location of cyclists in the bike box are shown in Fig. 5 and the waiting location of the first arriving cyclist is shown in Table 3. Comparing the waiting locations between cyclists going straight and those turning left at each intersection, shown in Table 4, we find a statistically significant effect with large effect sizes across the board. Cyclists going straight, especially the first arriving cyclist, mostly position themselves in location C and stay close to the curb. The second most common location is location D, on the approach to the bike box. This behaviour was especially observed when two or more cyclists were present in location C, blocking access to the bike box. Cyclists turning left, on the other hand, are more prone to wait in locations A and B. These cyclists position themselves in front of the waiting cars, in line with the design philosophy behind the bike box. In

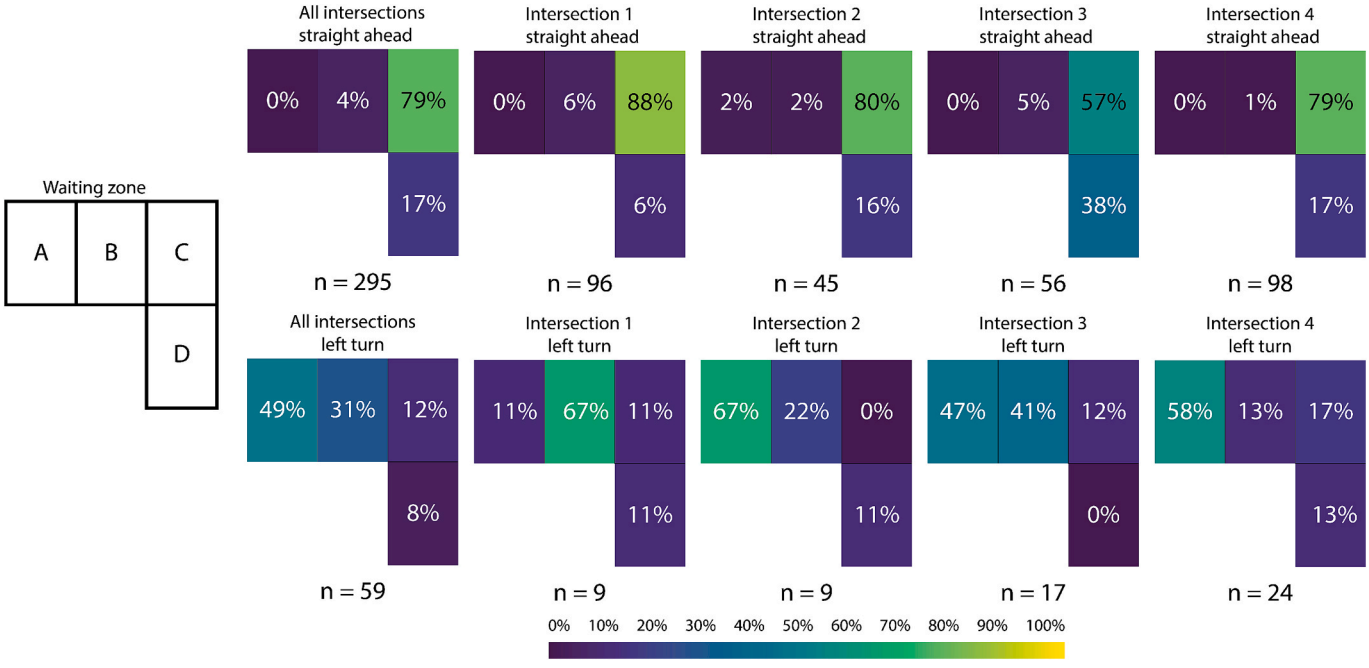


Fig. 5. Heatmaps showing the waiting locations of cyclists going straight and making a left turn at each intersection.

Table 3
Location of first arriving cyclist split out per direction.

	Intersection 1		*Intersection 2*		*Intersection 3*		*Intersection 4*	
Position	*Straight*	*Left*	*Straight*	*Left*	*Straight*	*Left*	*Straight*	*Left*
A	0 (0 %)	1 (13 %)	0 (0 %)	5 (83 %)	0 (0 %)	6 (43 %)	0 (0 %)	13 (62 %)
B	2 (3 %)	5 (63 %)	0 (0 %)	1 (17 %)	2 (7 %)	6 (43 %)	1 (1 %)	3 (14 %)
C	58 (95 %)	1 (13 %)	30 (100 %)	0 (0 %)	24 (80 %)	2 (15 %)	73 (99 %)	3 (14 %)
D	1 (2 %)	1 (13 %)	0 (0 %)	0 (0 %)	4 (13 %)	0 (0 %)	0 (13 %)	2 (10 %)

Table 4
Comparison between waiting locations when cyclists continue straight versus cyclists turning left using the Fisher exact test. Significant results are indicated with an asterisk. Significant results indicate that the waiting location of cyclists is different whether they continue straight or turn left. Effect sizes are small (<0.1), moderate (0.1 < 0.3), large (0.3 < 0.5), or very large (>0.5) as mentioned by Cohen (1988).

Location	**P-value**	**Effect size**
All	<2.2e-16**	0.782
Intersection 1	5.91e-08**	0.682
Intersection 2	1.77e-07**	0.825
Intersection 3	1.41e-11**	0.833
Intersection 4	7.22e-15**	0.795

* $p < 0.05$; ** significance level with Bonferroni correction $p < 0.010$ (0.05/5 = 0.010).

some cases, cyclists pass the queue of cars on the left-hand side, cycling against the flow of traffic in the opposite lane, to reach location A. This behaviour is mostly observed when other cyclists block their access to the bike box. It is also worth noting that we observed cyclists making use of the pavement on the left-hand side of the road to turn left at one of the intersections, bypassing the signal and bike box altogether. It is not clear whether this behaviour is due to red-light running or discomfort with waiting in the bike box.

Pairwise comparisons between the locations, shown in Table 5, show that the choice of waiting position is somewhat affected by the location of the measurement, albeit only when cyclists continue straight and with small effect sizes. The latter suggests that the actual differences are not large. The choice of where cyclists wait is statistically significantly different between intersection 3 and intersections 1 and 4. A key difference here is the number of cyclists waiting in Location D,. This can, to an extent, be governed by the design of the bike box, with the approach being less wide for intersection 3. Another possible reason for this effect is the relatively large number of cyclists occupying the bike box at the same time observed at intersection 3.

Table 5
Pairwise comparisons of locations using the Fisher exact test. Significant results are indicated with an asterisk. Significant results indicate that the choice of waiting zones is different between the compared intersections. Effect sizes (Cramer's V) are shown in brackets. Effect sizes are small (<0.1), moderate (0.1 < 0.3), large (0.3 < 0.5), or very large (>0.5) as mentioned by Cohen (1988).

Comparison	**Straight**	**Left**
Intersection 1 vs Intersection 2	9.84e-02 (0.210)	5.67e-02 (0.604)
Intersection 1 vs Intersection 3	4.12e-06** (0.395)	1.65e-01 (0.432)
Intersection 1 vs Intersection 4	1.84e-02* (0.205)	2.81e-01 (0.272)
Intersection 2 vs Intersection 3	1.86e-02* (0.284)	3.07e-01 (0.391)
Intersection 2 vs Intersection 4	4.64e-01 (0.133)	5.84e-01 (0.247)
Intersection 3 vs Intersection 4	1.09e-03** (0.284)	8.10e-01 (0.178)

* $p < 0.05$; ** significance level with Bonferroni correction $p < 0.008$ (0.05/6 = 0.008).

3.2. Questionnaire responses

3.2.1. Overview of questionnaire responses

An overview of the demographic information of the sample is shown in Fig. 6. The samples collected are slightly biased toward male respondents, making up 53 % of the total sample, 58 % of the sample in the Netherlands and 44 % of the sample in Sweden. The age groups are different between both countries, with most of the Dutch respondents in the age group of 18 to 24 (44 % of the respondents). For the Swedish sample, the age groups between 25 and 34 (33 %) and between 35 and 44 (30 %) form the majority. Most respondents in the Netherlands cycled daily (40 %) or at least three days a week (37 %), while the respondents in the Swedish sample cycled at least three days per week (38 %) or less than three days per week (35 %). Although Swedish respondents cycle less frequently, we can state that the sample is experienced with cycling.

When asked whether they understood the purpose of the bike box, most respondents stated to do so, 93 % of the respondents in the Netherlands and 85 % of the Swedish respondents. When prompted to explain the purpose of the bike box, participants from the Netherlands provided mixed responses. Common answers included references to improvements for the *safety* and *visibility* of cyclists, to provide *space for cyclists* and *reduce the risk of conflicts* when turning left, and to give cyclists a *head start*. The responses also included references to the bike box being intended to improve the (bicycle) traffic flow, or that it indicates that cyclists have priority over motorised traffic. Participants from Sweden provided similar explanations, correctly identifying that the bike box is intended to increase the visibility and safety of cyclists and provide a head start. In addition to these statements, some respondents indicate that the bike box is meant to *make left turns easier* for cyclists. As with the Dutch responses, we also find comments related to the bike box putting cyclists in a priority position over cars. In general, it can be stated that cyclists understand the purpose of the bike box, despite some answers suggesting that the bike box conveys a wrong message about its function, mainly that the bike box is meant to improve the flow of traffic and give cyclists priority.

When prompted whether they experience the bike box as safe, a majority (66 %) of the respondents in the Netherlands stated they feel safe when using the bike box. Respondents who answered that they did not feel safe in the bike box (34 %) were asked to state why. Key themes in these responses relate to feelings of *standing in the way of the cars* and *feeling rushed* or *pressured* by the cars behind them, feeling *unprotected* in the middle of the road, and a *lack of space* in the bike box. This suggests that, although the intention of the bike box is to provide cyclists with their own space to improve both their comfort and safety, the experience of cyclists does not always align with this intention. In Sweden, 25 % of respondents stated they feel unsafe when using the bike box. Here we again find statements related to *feeling rushed*. In addition to this, we find statements related to car drivers *not respecting the space for cyclists* and respondents *not trusting car drivers* in general. This suggests an influence of a feeling of mutual respect between road users on whether cyclists feel safe when standing in a bike box.

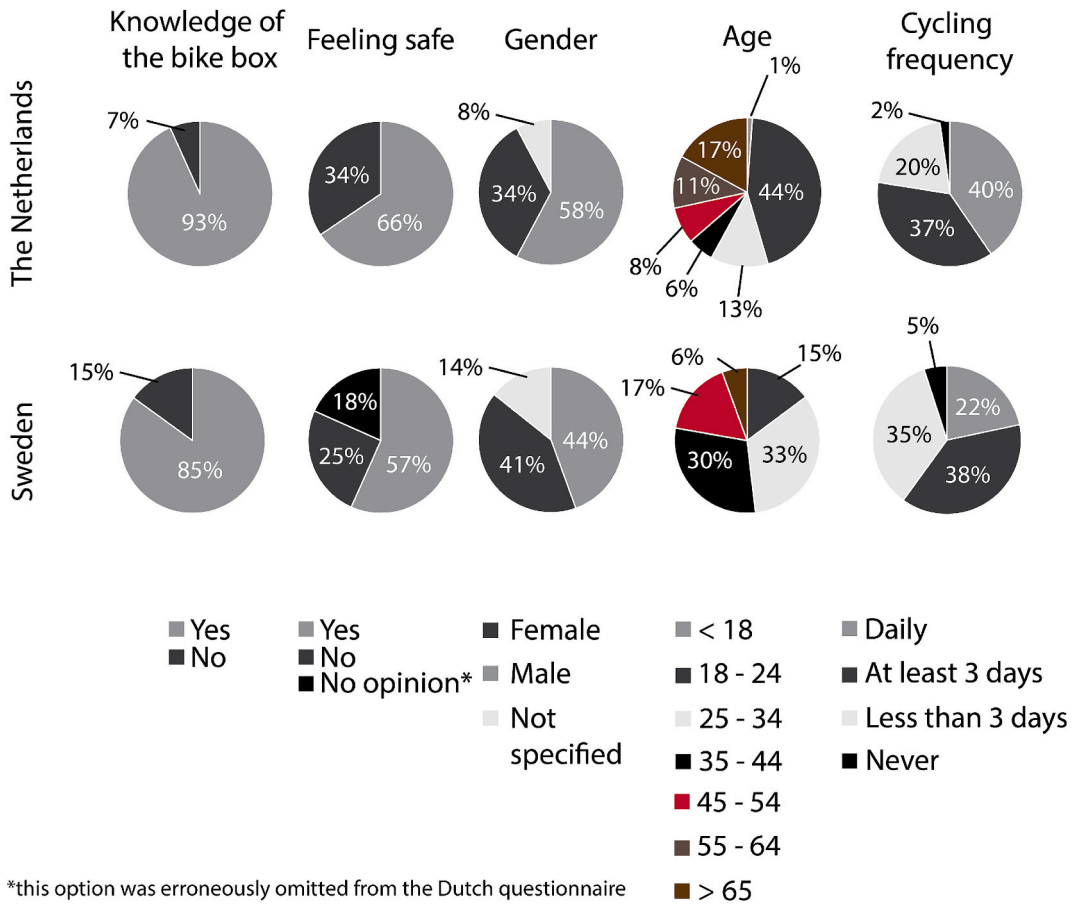


Fig. 6. Overview of the demographic variables for the sample from the Netherlands (top) and Sweden (bottom).

3.2.2. *Straight ahead versus making a left turn*

Fig. 7 shows a heatmap for the stated preference on where to wait in the bike box based on answers from the questionnaire. The choice of where to wait in the bike box is clearly affected by whether the cyclist continues straight ahead or makes a left turn. The difference in the frequency of the chosen waiting location as a function of turning direction is statistically significant regardless of the number of bikes in the bike box, as can be seen in Table 6, with large effect sizes for all cases. This is in line with the observed behaviour from the video observations and suggests that the stated preferences follow the actual behaviour shown in the bike box.

It is worth noting that the choice of waiting location is dependent on

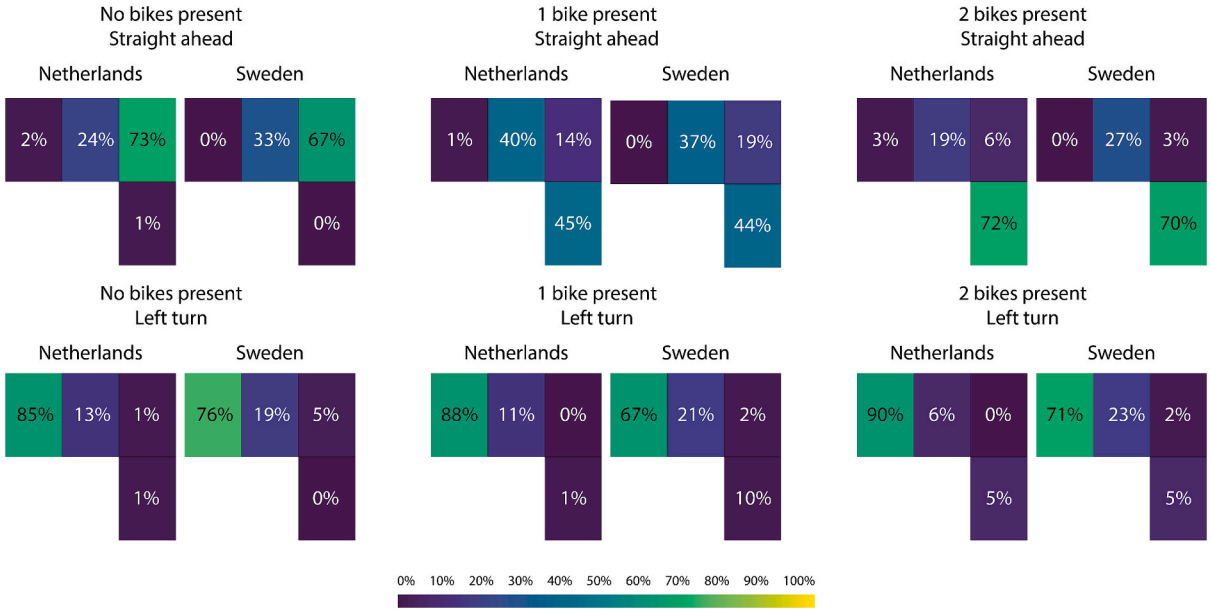


Fig. 7. Heatmaps showing the stated preference on where to stand in the bike box for the Netherlands and Sweden.

Table 6
Results of Fisher exact tests comparing waiting locations for both conditions (straight ahead versus left turn) in the Netherlands and Sweden. Significant results are indicated with an asterisk, showing a difference in the distribution of chosen waiting locations. Effect sizes (Cramer's V) are shown in brackets. Effect sizes are small (<0.1), moderate (0.1 < 0.3), large (0.3 < 0.5), or very large (>0.5) as mentioned by Cohen (1988).

Country	No bicycles in the bike box	1 bicycle in the bike box	2 bicycles in the bike box
The Netherlands	<2.2e-16** (0.871)	<2.2e-16** (0.882)	<2.2e-16** (0.872)
Sweden	<2.2e-16** (0.816)	<2.2e-16** (0.778)	<2.2e-16** (0.758)

* $p < 0.05$; ** Bonferroni correction $p < 0.0167$ (0.05/3 = 0.0167).

Table 7
Results of Fisher exact tests comparing waiting locations for different numbers of cyclists in the bike box in the Netherlands and Sweden. Significant results are indicated with an asterisk, showing a difference in the distribution of chosen waiting locations. Effect sizes (Cramer's V) are shown in brackets. Effect sizes are small (<0.1), moderate (0.1 < 0.3), large (0.3 < 0.5), or very large (>0.5) as mentioned by Cohen (1988).

	The Netherlands		*Sweden*	
Comparison	**Straight**	**Left**	**Straight**	**Left**
No bike vs One bike	8.53e-20** (0.659)	9.10e-01 (0.078)	2.27e-12** (0.596)	3.22e-01 (0.185)
No bike vs Two bikes	2.30e-30** (0.795)	1.46e-01 (0.171)	2.00e-23** (0.801)	4.95e-02* (0.243)
One bike vs Two bikes	1.00e-03** (0.294)	2.10e-01 (0.141)	3.09e-03** (0.303)	8.41e-01 (0.093)

* $p < 0.05$; ** significance level with Bonferroni correction $p < 0.0167$ (0.05/3 = 0.0167).

the number of cyclists when cyclists go straight, as shown in Table 7. When no other cyclists are present in the bike box, most respondents chose Zone C, with a few choosing Zone B. The most common reason for this choice is that cyclists want to *follow the bike lane* on the other side of the intersection, stating reasons such as "*no need to swerve right to get back into the bike lane*". Another common statement is related to cyclists *not wanting to block the cars* behind them, and that this location *feels the most secure*. Zone C is less common when one or two cyclists are present, indicating that the presence of other cyclists has an effect. Zone B is the preferred location to wait when one cyclist is present in the bike box, and Zone D when two cyclists are present in the bike box. Common reasons for choosing Zone B are the opportunity to *overtake the other cyclist* and to *stand next to the other cyclists*. Those who choose Zone D state to do so to *join the queue* of cyclists, often commenting that they do *not want to overtake* the other cyclists or are not sure of *the speed of the other cyclists*. Another common reason for choosing Zone D is that cyclists want to *avoid blocking cars*. Such statements go against the intention of the bike box. It is also worth noting that the effect size is largest when comparing the condition without a bike to the two other conditions. Suggesting that the presence of a bicycle has a strong effect on the preferred waiting choices of respondents, but this effect is less prominent when at least one bike is present.

When cyclists were prompted with questions related to making a left turn, most of the respondents answered Zone A in both the Netherlands and Sweden. Common statements for this are related to it being the *shortest route* when turning left, not having to cross *other lanes of traffic* and clearly *communicating their intention* to turn left. *Forcing other cars to stay behind* was another reason to choose Zone A. It is interesting to note that a common reason in Sweden is that this is where you are *supposed to stand*, a reason that was not present in the Dutch responses. Contrary to this are statements related to respondents choosing Zone A over the other zones to *avoid hindering cars* going straight ahead, which contradicts the idea behind the bike box. Such statements were present in both sets of answers and thus not related to the wider layout of the bike box in Sweden. It is worth noting that the popularity of Zone A did not depend on the presence of other cyclists in the bike box, as shown in Table 7. A similar trend in effect sizes as for the questions related to cyclists continuing straight is present here.

3.2.3. *The role of culture, gender, age, cycling habit, and feeling of safety*

Not all cyclists are the same, and it is possible that differences in culture, gender, age, cycling habits, and the feeling of safety affect where cyclists stand in the bike box. A summary of the results of these tests is shown in Table 8; for clarity reasons, we present the comparisons for the different age groups and cycling habits in the appendix. Although the study presented here is not comparative in nature, it is of interest to understand whether there is a difference between the responses in both countries. Such a difference would suggest an effect of (traffic) culture on the use of the bike box. Comparing the responses between respondents from both countries, we only find a statistically significant difference in the cases where a left turn would be made and two cyclists are present in the bike box, albeit with a small effect size. For all the other questions, we find no statistically significant differences, suggesting that there is no role of culture in the decision on where to stand in the bike box. This is especially relevant in light of the design differences, with the bike box spanning two lanes in Sweden. The absence of a difference in responses suggests that the wider bike box does not affect the preferred waiting location of cyclists.

Given the absence of a role of culture, we treat the whole sample for the tests concerning the role of gender, age, cycling habit, and feeling of safety. Gender or the feeling of safety when using a bike box is found to play no role in where respondents position themselves in the bike box, both when continuing straight and when turning left. Testing for differences in waiting locations based on the age of the respondents, Table A in the appendix, we find limited statistically significant differences. Noteworthy is that the biggest effects occur when continuing

Table 8
Summary of results of Fisher exact tests for different test conditions. Significant results are indicated with an asterisk, showing a difference in the distribution of chosen waiting locations. Effect sizes (Cramer's V) are shown in brackets. Effect sizes are small (<0.1), moderate (0.1 < 0.3), large (0.3 < 0.5), or very large (>0.5) as mentioned by Cohen (1988).

Cyclists going straight **Test condition**	**No bicycles in bike box**	**1 bicycle in bike box**	**2 bicycles in bike box**
Country (SE vs NL)	3.71e-01 (0.153)	8.05e-01 (0.100)	3.68e-01 (0.155)
Gender (female vs male)	8.45e-01 (0.098)	5.94e-01 (0.124)	9.32e-02 (0.212)
Feeling safe (yes vs no)	7.65e-01 (0.085)	6.33e-01 (0.124)	9.26e-01 (0.057)
Cyclists going left **Test condition**	**No bicycles in bike box**	**1 bicycle in bike box**	**2 bicycles in bike box**
Country (SE vs NL)	2.14e-01 (0.165)	3.74e-02 (0.222)	3.52e-03* (0.284)
Gender (female vs male)	3.63e-01 (0.153)	5.26e-01 (0.129)	1.30e-01 (0.193)
Feeling safe (yes vs no)	9.81e-02 (0.198)	1.68e-02 (0.255)	8.13e-02 (0.214)

*$p < 0.05$; ** Bonferroni correction $p < 0.0167$ (0.05/3 = 0.0167).

straight ahead and two cyclists are present in the bike box. In such cases, we see differences between the age groups of 45–54 with the age groups of 18–24 and 25–34. In both cases, we can observe that zone B, standing next to the cyclists, is chosen less often for the group of respondents between 45 and 54. The answers as to why this choice is made suggest that the older age group base their choice on their speed, stating answers related to *being slower* and *not wanting to overtake the other cyclists,* although no clear trend is present. Testing for an effect of the cycling habits of participants, shown in Table B in the appendix, we find no statistically significant difference in the choice of waiting zones. A possible reason for the lack of clear statistically significant differences between the age groups and cycling habits is the use of the Bonferroni correction. Relaxing this correction yields more statistically significant differences but risks Type I errors.

4. Discussion

4.1. Understanding and feeling of the bike box

With regards to cyclists understanding the intended purpose of the bike box, we find that most respondents state to understand its purpose and correctly describe the purpose of the bike box. This is similar to findings reported by Dill et al. (2012) in the USA, who found that a majority of their respondents understood the purpose of the bike box. It is worth noting that not all respondents provide a correct description of the purpose of the bike box, mentioning that it is intended to increase traffic flow and give priority to cyclists. These statements suggest that the intention of the bike box is not clear to all road users, which can lead to frustration and a wrong use of the infrastructure. This aligns with statements made by Christofa et al. (2023), who argued for the need for more education on the use of the bike box to increase its effectiveness. It is worth noting that although the bike box is also intended to reduce the exposure to exhaust fumes, no statements regarding this are made by the respondents.

When asked whether they feel safe when using the bike box, we find that a majority of cyclists experience the bike box as safe. This suggests that, for the most part, the bike box helps cyclists to feel safe when cycling through an intersection. Those who do not share this opinion state reasons related to feeling unprotected or being rushed by the cars behind them to cause an unsafe experience. In such cases, the envisioned head-start and space provided by the bike box are not experienced as safe by everyone. Furthermore, it suggests that cyclists experience a lack of mutual respect from car drivers, feeling rushed when having cars wait behind them. These findings suggest that the bike box alone is not sufficient to provide a feeling of safety for cyclists, and other types of interventions are necessary. Given the responses related to being rushed we argue that separating both flows, either physically or in time, would be the best avenue here.

4.2. Use of the bike box

Both the video observations and questionnaire responses highlight how the intended direction plays a major role in the use of the bike box, which is in line with findings presented by Kutusow (2022). We find that cyclists turning left mainly make use of Zone A, positioning themselves at the left-hand side of the bike box. Doing so means they use the bike box as intended, standing in front of the queue of cars. Cyclists continuing straight, however, are more likely to wait closer to the curb and in the approach lane of the bike box. This behaviour is dependent on whether there are other cyclists present in the bike box. When one or more cyclists are present, those who continue on straight opt to stand more in Zones B and D. This is not always caused by cyclists blocking access to the bike box, since this behaviour was observed with only a single cyclist present in the bike box as well. The questionnaire results show a similar effect, with respondents opting to wait behind the cyclist that is already present. Cyclists turning left are less affected by the number of cyclists waiting on the right-hand side of the bike box.

The desire for cyclists to continue straight to wait on the right-hand side of the bike box is also in line with previous work (c.f. de Lange, 2012; Dill et al., 2012; Scott-Deeter et al., 2023). These findings contradict statements made by Christofa et al. (2023), however, who found that when cyclists go straight they tend to position themselves in front of the cars. There is no clear indication to explain the difference in observations since the study by Christofa et al. (2023) only used video observations and did not question why cyclist chose their waiting location.

Although the four observed intersections differ in their design in terms of dimensions, presence of bike lanes, and size of the bike box we find limited differences with respect to where the cyclists position themselves in the bike box. The only statistically significant difference observed was related to cyclists going straight being more present in the approach lane for one intersection. A possible reason for this is the narrower approach lane, 1.5 versus 2 m, present at this intersection. This suggests that the use of the bike box is not strongly driven by the characteristics of an intersection itself, other than the width of the approach lane, and can be mostly attributed to the manoeuvre made by the cyclists.

Comparing both countries, based on the video observations and questionnaire responses, we find no statistically significant differences in the use of the bike box. This is with the exception of conditions where cyclists turn left, and two cyclists are already present in the bike box. These results suggest that the use of the bike box is not culturally driven, at least not between the observed samples in this study. Comparisons between gender, age, cycling habit, and feeling of safety did not show statistically significant differences in where cyclists stand in the bike box. This suggests that the behaviour we observe here is present across a wide array of cyclists and not limited to small sub-groups.

4.3. Motivation behind waiting location

The questionnaire responses highlight how the choice to wait on the right-hand side of the bike box when going straight is related to the presence of a cycle path on the opposite side of the intersection and cyclists wanting to join the queue when other cyclists are already present in the bike box. The effect of wanting to remain on the cycle path has also been hypothesised in different studies (Loskorn et al., 2013; Renčelj et al., 2019). It is worth noting that *not wanting to obstruct cars* was commonly mentioned as one of the reasons for cyclists to wait close to the curb when going straight. This behaviour goes against the intended purpose of the bike box, namely providing cyclists with a head-start and a dominant position in front of the queue of cars, and will also not yield the traffic calming effect described as one of the benefits of the bike box. Cyclists turning left indicate that their choice to wait more towards the left-hand side of the bike box is based on it being the shortest route, and indicate their intention to turn left to other road users.

4.4. Relevance for the implementation of bike boxes

Based on both the observations and the results from the questionnaire, it is clear that cyclists turning left (have the intention to) make proper use of the bike box in the majority of the cases. In reality, however, there seem to be very few cyclists turning left at an intersection, given the low number of left-turning cyclists at all four intersections studied here. Similar findings are reported by de Lange (2012) and by Kutusow (2022). The latter study observed cyclists at 3 intersections in Germany and found that cyclists continuing straight ahead are most common (between 75 % and 86 % of cyclists), and left-turning cyclists are less common (between 2 % and 12 % of cyclists in a bike box). This suggests that although the bike box is designed to accommodate left-turning cyclists, the most common movement by far is that of cyclists continuing straight. This group does not use the bike box as intended,

especially when other cyclists are already present. In those cases, the approach lane of the bike box is used to wait instead, effectively turning the bike box into a bike lane. As a result, the risk of blind spot crashes still exists since cyclists still wait next to the queue of cars.

Whilst it can be argued that cyclists standing close to the curb in Zone C are still in front of the queue of cars, it is still problematic. First, our results indicate that the likelihood of cyclists waiting in the approach lane increases as Zone C starts to fill. Furthermore, when cyclists position themselves close to the curb, they no longer slow down motorised traffic and reduce the effectiveness of the bike box in terms of providing a head start to cyclists. This behaviour is especially worrying as it is stated to be intentional, with respondents frequently stating they opt to wait next to the queue of cars in order to avoid obstructing them. Another worrying effect of this is that cyclists were observed to pass the queue of cars on the left-hand side to reach the bike box when the approach lane filled up. This is similar to the behaviour observed by de Lange (2012). Even though this results in a proper use of the bike box, it puts these cyclists at risk since they traverse the opposite lane for a period of time.

Based on these results, it is possible to state that bike boxes should be sparsely implemented on bicycle routes with a large number of cyclists continuing straight, despite the recommendations made in some design guidelines (CROW, 2016). In such cases, the bike box does not have the intended effect of placing cyclists in front of the queue of cars, and cyclists waiting in the approach lane will block access to the bike box for those who turn left. It is especially worth noting that our findings suggest that a low number of cyclists, as little as two cyclists even, is sufficient to block access to the bike box. This is a notable limitation of the bike box. With their access to the bike box blocked, cyclists who intend to make a left-hand turn end up either waiting in the bike lane, and thus have to cross the stream of cars when turning left, or bypass the queue of cars using the lane meant for traffic in the opposite direction. Neither is desirable from the perspective of increasing the safety of cyclists. It can be stated that the bike box provides a false sense of safety in these cases, and other avenues should be explored when the majority of cyclists continue straight. Alternatives such as protected intersections (Madsen and Lahrmann, 2017), are better suited in these conditions. When such a design is not feasible due to a lack of space, providing cyclists with their own signal phase, in line with the suggestions made by Casello et al. (2017), either alone or in combination with a bike box, can be beneficial to improve the safety of cyclists by actually separating cyclists from motorised traffic in time and space. Our results suggest that having a bike box without a separated signal phase does not solve the problem the bike box is intended to solve as conflicts between cyclists and motorised vehicles are still likely to arise. It is also worth noting that despite the four studied bike boxes complying with the guidelines in terms of their depth, access to the bike box was still blocked when cyclists positioned themselves to the right-hand side of the bike box. This suggests that the minimum depth suggested by the guidelines is not sufficient for the intended use of the bike box.

5. Limitations

One of the possible limitations of this research work is the lack of comparison of the questionnaire responses with the observations. One reason for this is the limited number of observations with either one or two cyclists present in the bike box. Furthermore, the observations were conducted separately from the questionnaire, meaning that those who were in the questionnaire and those we observed were not the same people. This is equal to assuming that the random selection of the respondent is representative of the cycling population for the two cities. Another limitation arises from the choice of location to observe. In all cases, the number of cyclists present during a single signal phase is relatively low. This means that the results presented here are mainly valid for conditions with a moderate to low number of cyclists, and we cannot make statements with regard to more heavily utilised conditions. On the other hand, a larger number of cyclists at the same time at the intersection can create the safety/density in number effect, which may hide/override the effect of the bike box. The final limitation is a result of the questionnaire design, where the other cyclists were always present on the right-hand side of the bike box. Our results can not indicate whether having cyclists already present on the left-hand side of the bike box would have an effect on where cyclists wait. It is also worth noting that the study presented here took place in two countries with a strong history of bicycle use and bicycle infrastructure. As such, the results presented here might not translate directly to more car-centric countries where the familiarity with infrastructure and presence of cyclists is lower. That being said, we find similar behaviour for cyclists going straight as Dill et al. (2012) did in their case study in the USA, presenting a more car-centric condition, suggesting that the effect of cycling culture might be limited, and the use of the bike box is not necessarily driven by mobility culture.

6. Conclusion

The study presented here focuses on the use of the bike box in the Netherlands and Sweden, two countries with a significant modal share of cyclists and a history of applying the bike box at signalised intersections. For the bike box to work as intended and make cyclists more visible, reducing the risk of blind spot conflicts, it is important that cyclists make proper use of the bike box. The results presented here suggest that this is not the case. Although cyclists turning left were observed to generally use the bike box as intended, and the questionnaire responses revealed that the intention to do so is present, this group of cyclists is relatively uncommon. The majority of the cyclists at the four observed intersections were observed to continue straight ahead. These cyclists tend to stay to the right-hand side, blocking access to the bike box, and later arriving cyclists make use of the approach lane to wait. As a result, the bike box is effectively used as a bike lane. This behaviour was reported in the questionnaire responses, with comments made about not wanting to block the queue of cars and wanting to follow the bike lane as a reason for this choice, as well as observed in the video observations. As such, it is likely that the envisioned safety effects are not achieved. The study presented here is limited to where and why cyclists position themselves in the bike boxes, and the connection with safety is indirect or related to other literature studies. Future studies could include studies into the interaction between cyclists and motorists through the use of conflict analysis techniques, comparing intersections with and without bike boxes (Johnsson et al., 2021). Such a conflict analysis should also account for conflicts between cyclists who make use of the bike box. Another avenue of further research is to better understand why left-turning cyclists are not frequent at intersections, which may depend on the perception of safety (Yastremska-Kravchenko et al., 2022), the manoeuvre and where the bike boxes can make the biggest differences on how the road infrastructure is used. Our findings also highlight that the minimum depth of the bike box is potentially not sufficient to allow enough space for cyclists the manoeuvre into the bike box between cyclists already waiting and the first car in the queue. A future research direction is to better understand what the minimum design criteria for the bike box should be in terms of its depth and approach lane design, also considering the maximum number of cyclists that can access and use the space provided by the bike box. The latter is relevant as our

findings suggest that a small number of cyclists is already sufficient to block access to the bike box. Finally, the results presented here might not be applicable to more car-centric conditions, as such more work should be done on understanding the behaviour of cyclists in countries with a dominant modal share of cars.

CRediT authorship contribution statement

Ruben Alaric Kuipers: Writing – review & editing, Writing – original draft, Visualization, Software, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Carmelo D'Agostino:** Writing – review & editing, Supervision, Funding acquisition, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A

Table A
Summary of results of Fisher exact tests for different age groups. Significant results are indicated with an asterisk showing a difference in the distribution of chosen waiting locations. Effect sizes (Cramer's V) shown in brackets. Effect sizes are small (<0.1), moderate (0.1 < 0.3), large (0.3 < 0.5), or very large (>0.5) as mentioned by Cohen (1988).

Test group	No bicycles in bike box (straight)	1 bicycle in bike box (straight)	2 bicycles in bike box (straight)	No bicycles in bike box (left)	1 bicycle in bike box (left)	2 bicycles in bike box (left)
18–24 vs 25–34	6.71e-01 (0.127)	1.42e-01 (0.224)	5.02e-01 (0.186)	7.28e-01 (0.157)	6.91e-01 (0.150)	5.25e-01 (0.185)
18–24 vs 35–44	1.00e+00 (0.084)	6.09e-01 (0.116)	1.00e+00 (0.084)	1.73e-01 (0.260)	5.26e-02 (0.276)	1.78e-02* (0.329)
18–24 vs 45–54	1.00e+00 (0.089)	1.97e-01 (0.234)	1.93e-03** (0.469)	1.00e+00 (0.110)	1.00e+00 (0.087)	2.40e-01 (0.222)
18–24 vs 55–64	5.53e-01 (0.229)	4.19e-01 (0.228)	1.81e-01 (0.265)	7.67e-01 (0.125)	5.68e-01 (0.116)	3.30e-01 (0.184)
18–24 vs > 65	1.00e+00 (0.089)	4.33e-01 (0.154)	6.34e-01 (0.096)	5.58e-01 (0.227)	2.23e-01 (0.227)	8.23e-01 (0.143)
25–34 vs 35–44	5.58e-01 (0.099)	3.60e-01 (0.215)	7.38e-01 (0.151)	3.17e-01 (0.244)	1.25e-01 (0.310)	7.11e-02 (0.348)
25–34 vs 45–54	5.17e-01 (0.135)	3.46e-01 (0.217)	1.70e-03 ** (0.513)	5.90e-01 (0.178)	1.00e+00 (0.163)	8.62e-01 (0.186)
25–34 vs 55–64	3.80e-01 (0.226)	4.71e-01 (0.251)	6.06e-01 (0.222)	1.00e+00 (0.057)	9.02e-01 (0.145)	8.00e-01 (0.177)
25–34 vs > 65	5.17e-01 (0.135)	1.00e+00 (0.067)	6.01e-01 (0.232)	1.18e-01 (0.315)	3.71e-01 (0.288)	5.94e-01 (0.234)
35–44 vs 45–54	1.00e+00 (0.043)	7.60e-01 (0.14)	6.37e-03* (0.516)	1.48e-01 (0.252)	7.00e-02 (0.347)	2.15e-01 (0.315)
35–44 vs 55–64	6.88e-01 (0.193)	8.01e-01 (0.211)	2.53e-01 (0.308)	2.48e-01 (0.294)	4.78e-01 (0.21)	5.36e-01 (0.191)
35–44 vs > 65	1.00e+00 (0.043)	5.62e-01 (0.193)	8.32e-01 (0.186)	1.08e-02* (0.428)	2.30e-03* (0.501)	1.50e-02* (0.436)
45–54 vs 55–64	8.38e-01 (0.186)	5.81e-01 (0.263)	5.39e-02 (0.467)	7.94e-01 (0.184)	4.84e-01 (0.214)	8.22e-01 (0.185)
45–54 vs > 65	1.00e+00 (0.000)	3.75e-01 (0.247)	8.67e-02 (0.447)	4.85e-01 (0.25)	4.85e-01 (0.250)	2.27e-01 (0.349)
55–64 vs > 65	8.38e-01 (0.186)	8.49e-01 (0.216)	7.71e-01 (0.214)	1.03e-01 (0.326)	4.45e-02* (0.383)	1.63e-01 (0.328)

* $p < 0.05$; ** Bonferonni correction $p < 0.003$ (0.05/15 = 0.003).

Table B
Summary of results of Fisher exact tests for different cycling habits. Significant results are indicated with an asterisk showing a difference in the distribution of chosen waiting locations. Effect sizes (Cramer's V) shown in brackets. Effect sizes are small (<0.1), moderate (0.1 < 0.3), large (0.3 < 0.5), or very large (>0.5) as mentioned by Cohen (1988).

Test group	No bicycles in bike box (straight)	1 bicycle in bike box (straight)	2 bicycles in bike box (straight)	No bicycles in bike box (left)	1 bicycle in bike box (left)	2 bicycles in bike box (left)
At least 3 days per week vs Daily	1.00e+00 (0.099)	1.53e-01 (0.209)	4.88e-01 (0.175)	2.91e-02* (0.259)	1.45e-01 (0.194)	6.03e-01 (0.116)
At least 3 days per week vs Less than 3 days per week	5.10e-01 (0.138)	9.77e-01 (0.094)	3.10e-01 (0.189)	1.00e+00 (0.089)	7.88e-01 (0.080)	8,06E-01 (0,085)
At least 3 days per week vs Never	5.98e-01 (0.107)	1.00e+00 (0.139)	1.00e+00 (0.053)	2.61e-01 (0.267)	2.48e-01 (0.431)	1.04e-01 (0.439)
Daily vs Less than 3 days per week	5.42e-01 (0.170)	1.34e-01 (0.219)	9.73e-01 (0.034)	1.43e-02* (0.269)	3.95e-02* (0.248)	2.44e-01 (0.166)
Daily vs Never	6.50e-01 (0.095)	5.49e-01 (0.187)	1.00e+00 (0.096)	6.21e-02 (0.350)	3.47e-02* (0.451)	6.74e-02 (0.465)
Less than 3 days per week vs Never	7.03e-01 (0.076)	7.25e-01 (0.138)	1.00e+00 (0.098)	2.77e-01 (0.284)	1.07e-01 (0.436)	1.26e-01 (0.441)

* $p < 0.05$; ** Bonferonni correction $p < 0.008$ (0.05/6 = 0.008).

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